When Reconstruction Passes: Natural-Language Autoencoders on VLA Activations Fail Grounding and Semantic Steering

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Abstract

Natural-language autoencoders (NLAs) map neural activations to text and back, offering readable interfaces to model internals. We port the NLA recipe to the GR00T vision-language-action (VLA) backbone: extract per-token hidden states, train an activation verbalizer (AV) and reconstructor (AR), and inject AR outputs as live backbone steers in LIBERO simulation. Supervision uses a multimodal *teacher* (frames + instruction), not human descriptions of what **h** encodes—a confound inherited from the original LLM setting. We release `nla-groot` and a three-axis evaluation protocol: (1) offline reconstruction and retrieval, (2) vision-grounded caption judging, (3) closed-loop steer A/B (matched vs. mismatched language). On our main checkpoint, reconstruction metrics pass while grounding, anti-template specificity, and semantic steering fail; aggregate scores **hide** collapse on `image_patch` tokens, where retrieval margin is near zero. **Claim:** NL autoencoders on VLA activations are misleading without vision-grounded and behavioral checks *stratified by token role*—use this protocol before trusting captions or steering policies with language.

1 Introduction

Vision-language-action policies such as GR00T [1] couple a multimodal backbone to a diffusion action head. The backbone hidden state $\mathbf{h} \in \mathbb{R}^{2048}$ at layer 16 is causally upstream of actions, yet there is no standard way to read or write **h** in natural language. Fraser-Taliente et al. [2] propose *natural-language autoencoders*: AV(**h**) produces a caption and AR(y) reconstructs $\hat{\mathbf{h}} \approx \mathbf{h}$, yielding both an interpretability artifact and a causal handle for activation steering [3, 4].

We ask whether this recipe *transfers* to a VLA backbone, where (i) activations are multimodal and task-conditioned, (ii) downstream *behavior* is the right falsification test, and (iii) training labels are **synthetic**: a GPT-class teacher sees camera frames and token metadata, *not* raw **h** floats. The teacher may describe scene attributes only weakly present in **h**; supervised fine-tuning (SFT) therefore optimizes $P(\text{teacher text} \mid \mathbf{h})$, not faithful "what **h** means."

Contribution. We do not claim a strong AV or working semantic steer. We claim a **negative, actionable result**: autoencoder metrics can look successful while captions are vision-ungrounded and steering is behaviorally non-semantic. We release open tooling and a pre-registered **three-axis scorecard** so NLA ports on policies are not validated by reconstruction alone.

2 Setup (minimal)

Activations. A forward hook on GR00T `backbone_features` (Qwen3-VL, layer 16, pre-action-head LayerNorm) records $\mathbf{h}_i$ per token at three position types: `last_text`, `image_patch`, `anchor`. Types are inferred from the backbone's `image_mask` (`input_ids == image_token_id`) and `attention_mask`: `image_patch` = valid vision placeholders; `last_text` = last valid non-image token; `anchor` = last valid token overall (often the sequence tail). Corpus: LIBERO Goal/Spatial/Object/10 LeRobot trajectories; $\alpha \approx \text{P75}\|\mathbf{h}\|_2$ for injection scaling.

AV **and** AR**.** Both are LoRA [5] fine-tunes of Qwen3-4B-Instruct. AV injects $\alpha \cdot \text{normalize}(\mathbf{W}_p\mathbf{h})$ at a reserved slot token and is trained with CE on teacher bullets. AR reads "Summary of the following text:" and predicts $\hat{\mathbf{h}}/\alpha$; loss is MSE plus InfoNCE hard negatives. Joint SFT (`libero_4suite_v3`: 15k steps, batch 4, optional `ar_av_mix` to expose AR to AV samples).

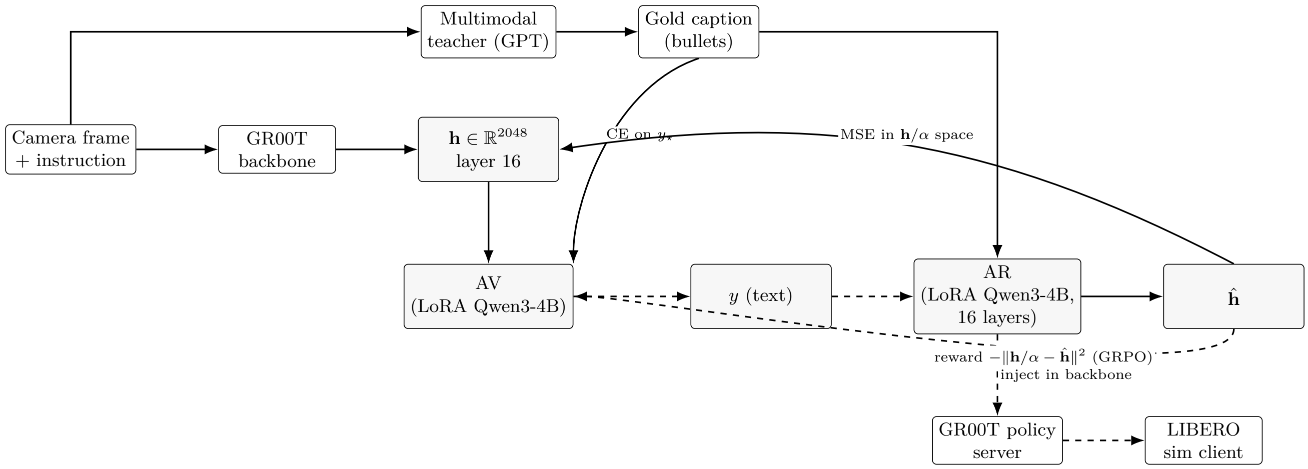

Figure 1: **Pipeline.** Solid: training ($\mathbf{h}$ from GR00T hook; gold captions from teacher). Dashed: inference ($\hat{\mathbf{h}} = \mathrm{AR}(y)$ steers the live policy). Auditing uses $\mathrm{AV}(\mathbf{h})$; steering bypasses AV.

Steering. At inference, user text y maps to $\hat{\mathbf{h}} = \mathrm{AR}(y)$ and blends into backbone tokens (`image_patch` or `image_patch_all`) on every `get_action` call in the official LIBERO policy server.

3 Three-axis evaluation protocol

Each checkpoint is scored on **independent** axes; overall PASS requires required metrics (pre-registered in `build_v3_scorecard.py`).

Axis 1: Reconstruction & retrieval. Closed-loop cosine $\cos(\mathbf{h}, \mathrm{AR}(\mathrm{AV}(\mathbf{h})))$ without teacher forcing; retrieval margin (matched vs. cross-pair $\mathrm{AR}(\mathrm{AV}(\mathbf{h}))$). Report both **aggregate** and **per-`position_type`** slices (`image_patch` vs. `last_text` vs. `anchor`): the pooled margin can pass while `image_patch` is near chance. High scores mean AV and AR form a consistent codec—not that language matches pixels.

Axis 2: Vision-grounded judge. `llm_judge_av_captions.py` scores captions against *cached frames* (grounding / specificity; appropriateness for position type). We report gold (teacher) vs. AV greedy on identical rows; gold sets a ceiling ($\sim$75–83% grounding on held-out samples).

Axis 3: Closed-loop steer A/B. On `put_the_bowl_on_the_plate` (LIBERO Goal, GR00T checkpoint, seeds $\{0, 1, 2\}$): baseline, *correct* steer text, *wrong* text. Primary metric: $\Delta_{cw} = \mathrm{succ}_{\mathrm{correct}} - \mathrm{succ}_{\mathrm{wrong}}$ (PASS if $\geq$ 5pp). Semantic steering requires $\Delta_{cw} > 0$, not merely $\mathrm{succ} \neq \mathrm{succ}_{\mathrm{baseline}}$.

Auditing (optional fourth arm). Counterfactual panel: baseline / edited / control AV explanations under $\mathbf{h}$ edits (`run_interp_panel.py`) for hypothesis testing; complements Axes 2–3.

4 Results: metrics pass, semantics fail

Table 1 is the V3 scorecard (**overall FAIL**). Retrieval margin **passes** (0.124 vs. 0.05 threshold); judge anti-template specificity **fails** (8.3% vs. 50%); $\Delta_{cw} = 0$ (WARN).

Template collapse (Axis 2). Table 3: on 4-suite holdout, gold grounding is 75% but AV is 41.7%; anti-template specificity drops from 50% (gold) to **8.3%** (AV)—captions collapse to reusable scene templates (“kitchen counter / green bowl”) across different tasks. Appropriateness stays 100% for AV: prose *looks* valid while being vision-wrong (*shorthand collapse*).

Table 1: V3 scorecard (`libero_4suite_v3`). Required metrics gate overall verdict.

Metric	Value	Thr.	Verdict
retrieval_margin	0.124	0.05	PASS
judge_grounding_specific	0.417	0.55	WARN
judge_anti_template	0.083	0.50	FAIL
sim_correct_minus_wrong	0.000	0.05	WARN

Table 2: Stratified V3 metrics (`libero_4suite_v3`). Closed-loop cosine = $\cos(\mathbf{h}, \mathrm{AR}(\mathrm{AV}(\mathbf{h})))$ greedy; margin from `closed_loop_retrieval.py`. Judge B% = grounding-specific pass rate on a 32-row held-out subsample (8 per type targeted; see text).

Position	n	CL cos	Margin	AV B%
`image_patch`	85	0.338	0.003	0
`last_text`	85	0.437	0.046	67
`anchor`	8	$0.367^{\dagger}$	0.139	75
pooled	178	0.388	0.124	42

†Teacher-forced AR cosine on gold (no closed-loop `anchor` slice in val log).

Aggregate metrics hide token-role failure (Axes 1–2). Table 2: the **pooled** retrieval margin passes (0.124), but the `image_patch` stratum is near chance (margin 0.003, n=85): AV–AR is not discriminative on vision tokens—the regime where NLAs are meant to add value over text-only readouts. `last_text` rows look healthier (closed-loop cosine 0.44 vs. 0.34; judge grounding 67% vs. 0% on the small per-type judge subsample), consistent with instruction paraphrase rather than pixel grounding. `anchor` retrieval margin is high but n=8 (saturated @5); do not over-interpret. **Lesson:** headline scorecards on mixed position types can certify a codec that works on language slots while vision slots collapse.

Symmetric steering (Axis 3). Baseline solves all seeds (succ = 1.0). Every steer arm fails (succ = 0) for both *matching* and *contradictory* prompts—$\Delta_{cw} = 0$. Bowl displacement ≈0.145 m (baseline) vs. ≈0.07 m (all steers): intervention *dampens* motion uniformly rather than redirecting toward the prompted object. **Steerability ≠ faithful interpretability.**

Negative result in one line. Reconstruction and retrieval can certify a low-loss text–vector codec while (i) captions fail vision grounding (especially on `image_patch`), and (ii) language steers do not separate correct from wrong instructions. Aggregate passes are insufficient—stratify by token role.

5 Use this before you trust or steer

For auditing. Do not treat AV($\mathbf{h}$) as ground truth without Axis 2 on held-out frames and, when possible, counterfactual $\mathbf{h}$ edits (Axis 4). Report gold-vs-AV gaps *per `position_type`*, not pooled AV alone; `image_patch` is the vision-grounding slice.

For steering. Do not deploy AR(y) backbone steers because reconstruction is high. Require $\Delta_{cw} > 0$ on held-out tasks and dose calibration; until then, treat NL steers as *unvalidated interventions*.

For training. Teacher-only SFT lacks an explicit pixel alignment term; reconstruction rewards template IDs invertible by AR. Mitigations we document but do not fully solve here: label QA, quality-weighted SFT, mined hard negatives, judge-blended GRPO.

Reproducibility. Anonymous code: https://anonymous.4open.science/ (placeholder—replace before submit). Scorecard: `scripts/eval/build_v3_scorecard.py` (includes `by_position_type` breakdown); commands: `paper/repro/canonical_commands.sh`.

Table 3: Multimodal judge pass rates (24-row held-out sample).

Checkpoint	Source	Ground.	Anti-templ.
v3 (4-suite)	gold	0.750	0.500
	AV	0.417	0.083

Broader impact

NL interfaces to policy internals could help safety auditing or harm if misread as faithful; our protocol is meant to reduce overclaim.

References

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